APPROVED ASSEMBLIES

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: 18-22 ga. Steel

System Type C(1): Minimum 22 ga., Type B, Grade 33 steel deck fastened with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support, spaced maximum 6-feet on center. Deck side laps secured 30 inches on center with ITW Buildex Traxx/1 fasteners.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft²
InsulBase, SecuShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.4" thick	**1, 2, 4, 5**	**1:3.2 ft²**
Minimum 2" thick	**1, 2, 6, 7, 8, 9**	**1:4 ft²**
Minimum 2" thick	**1,2 with 5**	**1:2 ft²**
Dens Deck, Dens Deck Prime		
Minimum ¼" thick	**1, 2, 6, 7, 8, 9**	**1:2 ft²**

Note: All layers shall be simultaneously fastened; see top layer below for fasteners and density. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment. Single and multiple layers of insulation can be attached to base layer with Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank Carlisle OlyBond 500BA, or Carlisle One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane #1: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.
Maximum Design Pressure –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 gage, Type B, ASTM A 653 SS Grade 33 Steel deck fastened to steel support at a maximum span of 6 feet o.c. Steel deck shall be fastened with minimum ITW Buildex Traxx/4 at a maximum spacing of 6 inches o.c. Side laps shall be fastened with ITW Buildex Traxx/1 at a maximum spacing of 24 inches o.c.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(2): All layers of insulation simultaneously attached; membrane fully adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
One of the following covered with the boards listed in Top Layer Top Layer.		
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield HD Composite **Minimum 1.5" thick**	**N/A**	**N/A**
Dens Deck, Dens Deck Prime **Minimum ¼" thick**	**N/A**	**N/A**

Note: All layers shall be simultaneously fastened; see top layer below for fasteners and density. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment. Single and multiple layers of insulation can be attached to base layer with Carlisle Flexible FAST Adhesive, Carlisle OlyBond 500BA, or Carlisle One-Step Adhesive.

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
Required over the insulations listed in Base Insulation Layer: **Plywood** **Minimum 19/32" thick**	**1, 2**	**1:1.9 ft²**

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 gage (min 0.0294in.) ASTM A 653 SS Grade 33 Steel deck fastened to steel support at a maximum span of 6 feet o.c. Steel deck shall be fastened with minimum ITW Buildex Traxx/4 at a maximum spacing of 6 inches o.c. Side laps shall be fastened with ITW Buildex Traxx/1 at a maximum spacing of 24 inches o.c.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(3): All layers of insulation simultaneously attached; membrane fully adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft²
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Comoposite		
Minimum 2" thick	**1, 2, 6, 7, 8, 9**	**1:1.6 ft²**

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment. Single and multiple layers of insulation can be attached to base layer with Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank Adhesives.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane #1: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld..
Maximum Design Pressure –75 psf (See General Limitation #7)

Membrane #2: Sure-Weld AFX membrane adhered to the insulation using SureMB MC Adhesive. The adhesive is applied to the substrate only at a rate of 67 ft²/gal. 1 gal/67 ft.² Outside 1.5-inch of side laps are heat welded. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.
Maximum Design Pressure –60 psf. (See General Limitation #7)

Maximum Design Pressure: See Membrane Options Above

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 gage ASTM A 611 Grade E or A653 Grade 80 Steel deck fastened to steel support at a maximum span of 6 feet o.c. Steel deck shall be fastened with minimum ITW Buildex Traxx/5 at a maximum spacing of 6 inches o.c. Side laps shall be fastened with ITW Buildex Traxx/1 at a maximum spacing of 24 inches o.c.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(4): All layers of insulation simultaneously attached; membrane fully adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
One of the following covered with the boards listed in Top Layer.		
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5" thick	**N/A**	**N/A**

Note: All layers shall be simultaneously fastened; see top layer below for fasteners and density. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment. Single and multiple layers of insulation can be attached to base layer with Carlisle Flexible FAST Adhesive.

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Required over the insulations listed in Base Layer.		
Dens Deck Prime (For use over all insulation types.) Minimum ⅝" thick	**5**	**1:1.33 ft²**

Note: <u>All layers shall be simultaneously fastened; see top layer for fasteners and density</u> Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

System Type C(5): All layers of insulation simultaneously attached (optional layers adhered); membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield , SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**1, 2 with 5**	**1:2 ft²**

Optional layers of insulation may be attached to the base layer with Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank Adhesives applied to the base layer in approximately ¾ to 1-inch wide beads spaced 12-inches on center.

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with , Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 12-inches on center. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

System Type C(6): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft^2**
InsulBase, SecurShield,, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft^2**
Dens Deck Prime Minimum 1/4-inch thick	**1, 2 with 5**	**1:2 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 12-inches on center. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

System Type C(7): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	1, 2 with 5	1:3.2 ft²

Optional layers of insulation may be attached to the base layer with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank. Adhesives applied to the base layer in approximately ¾ to 1-inch wide beads spaced 12-inches on center.

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation using, Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate at a rate of 1 gal/sq., full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf. (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

System Type C(8): Base layer(s) of insulation fastened, top layer adhered; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield,, SecurShield HD Composite,, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**1, 2 with 5**	**1:3.2 ft²**

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Top layer of insulation is attached to the base with Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank.l Adhesives applied to the base layer in approximately ¾ to 1-inch wide beads spaced 12-inches on center.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation using Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate at a rate of 1 gal/sq., full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf. (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

System Type C(9): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield,, SecurShield HD Composite, H-Shield,.H-Shield CG , H-Shield HD Composite Minimum 2-inch thick	**1, 2 with 5**	**1:4 ft²**

Optional layers of insulation may be attached to the base layer with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank. Adhesives applied to the base layer in approximately ¾ to 1-inch wide beads spaced 12-inches on center.

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation using Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate at a rate of 1 gal/sq., full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf. (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

System Type C(10): Base layer(s) of insulation fastened, top layer adhered; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield , SecurShield HD Composite , H-Shield , H-Shield CG, H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:4 ft²**

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Top layer of insulation is attached to the base with Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank. Adhesives applied to the base layer in approximately ¾ to 1-inch wide beads spaced 12-inches on center.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate at a rate of 1 gal/sq., full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf. (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(11): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield , SecurShield HD Composite , H-Shield , H-Shield CG , H-Shield HD Composite		
Minimum 1.5-inch thick	**1, 2 with 5**	**1:1.6 ft²**

Optional layers of insulation may be attached to the base layer with Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank. Adhesives applied to the base layer in approximately ¾ to 1-inch wide beads spaced 6-inches on center.

Note: Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation using Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate at a rate of 1 gal/sq., full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –82.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(12): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite, H-Shield CG , H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Minimum 60-mil Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive . The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.
OR
Sure-Weld FleeceBACK membrane adhered to the insulation with , Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 4-inches on center or applied in a full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –82.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 69-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(13): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite, H-Shield CG , H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Minimum 60-mil Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive . The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –90 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 66-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(14): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite, H-Shield, H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Minimum 60-mil Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive . The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –97.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 64-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(15): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite, H-Shield CG, H-Shield HD Composite. Minimum 2-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Minimum 60-mil Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –105 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 62-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(16): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite , H-Shield CG , H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Minimum 60-mil Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive . The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –112.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(17): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite , H-Shield CG , H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Minimum 60-mil Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive . The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –120 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 62-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.
Or
Minimum 20 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 69-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.
Or
Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(18): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite , H-Shield CG, H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 4-inches on center or applied in a full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –112.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleecBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(19): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite, H-Shield CG, H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 4-inches on center or applied in a full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –135 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(20): Base layer(s) of insulation fastened, top layer adhered; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield , SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	**1, 2, 4 with 5**	**1:1.8 ft²**

Note: The base layer of insulation is fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**

Note: Top insulation layer is adhered with Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug applied in ¾ to 1-inch wide beads spaced 6-inches on center.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –52.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(21): Base layer(s) of insulation fastened, top layer adhered; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft²
InsulBase, SecurShield,, SecurShield HD Composite, H-Shield, H-Shield CG , H-Shield HD Composite Minimum 1.5-inch thick	**1, 2, 4 with 5**	**1:1.33 ft²**

Note: The base layer of insulation is fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft²
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**

Note: Top insulation layer is adhered with Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug applied in ¾ to 1-inch wide beads spaced 4-inches on center.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –82.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

System Type C(22): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG , H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
SECUROCK Gypsum-Fiber Roof Board		
Minimum 1/2-inch thick	**1, 2 with 5**	**1:3.2 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.
OR
Sure-Weld FleeceBACK membrane adhered to the insulation with Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 4-inches on center or applied in a full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld. (

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

System Type C(23): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
InsulBase, SecurShield , SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
SECUROCK Gypsum-Fiber Roof Board		
Minimum 5/8-inch thick	**1, 2 with 5**	**1:4 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.
OR
Sure-Weld FleeceBACK membrane adhered to the insulation Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 4-inches on center or applied in a full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(24): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.0-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
SECUROCK Gypsum-Fiber Roof Board Minimum ½-inch thick	**1, 2 with 5**	**1:1.66 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.
OR
Sure-Weld FleeceBACK membrane adhered to the insulation Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 4-inches on center or applied in a full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –75 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(25): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
SECUROCK Gypsum-Fiber Roof Board Minimum ½-inch thick	**1, 2 with 5**	**1:1.66 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.
OR
Sure-Weld FleeceBACK membrane adhered to the insulation with Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 4-inches on center or applied in a full coverage. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –82.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 20 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 67-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(26): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, Minimum 1.5-inch thick	**H-Shield CG, H-Shield HD Composite N/A**	**N/A**

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SECUROCK Gypsum-Fiber Roof Board Minimum ½-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –97.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 20 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 65-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 58-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(27): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SECUROCK Gypsum-Fiber Roof Board		
Minimum ½-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 20 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 65-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 58-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(28): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SECUROCK Gypsum-Fiber Roof Board Minimum ½-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 20 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 63-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 57-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners spaced 6 inches on center along each structural support. Deck side laps secured 24 inches on center with ITW Buildex Traxx/1 fasteners.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(29): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield , SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SECUROCK Gypsum-Fiber Roof Board		
Minimum ½-inch thick	**1, 2 with 5**	**1:1 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(30): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield,, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:4 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(31): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG , H-Shield HD Composite		
Minimum 1.5-inch thick	**1, 2 with 5**	**1:2 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(32): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**1, 2 with 5**	**1:3.2 ft^2**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft^2/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(33): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft²
InsulBase, H-Shield Minimum 2-inch thick	**1, 2, 4 with 5**	**1:1.66 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –60 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(34): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SecurShield, SecurShield HD Composite, H-Shield CG, H-Shield HD Composite		
Minimum 2-inch thick	**1, 2 with 5**	**1:2 ft²**

Note: All layers shall be simultaneously fastened. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –60 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(35): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
SECUROCK Gypsum-Fiber Roof Board Minimum ¼-inch thick	**1, 2 with 5**	**1:2 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(36): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft^2**
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft^2**
SECUROCK Gypsum-Fiber Roof Board Minimum ½-inch thick	**1, 2 with 5**	**1:3.2 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(37): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG , H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
SECUROCK Gypsum-Fiber Roof Board Minimum 5/8-inch thick	**1, 2 with 5**	**1:4 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(38): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG , H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
SECUROCK Gypsum-Fiber Roof Board Minimum 5/8-inch thick	**1, 2 with 5**	**1:4 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(39): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft^2**
InsulBase, SecurShield , SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft^2**
SECUROCK Gypsum-Fiber Roof Board		
Minimum ½-inch thick	**1, 2, 4 with 5**	**1:1.66 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –60 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(40): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
Dens Deck Prime Minimum ½-inch thick	**1, 2, 4 with 5**	**1:3.2 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

System Type C(41): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.5-inch thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
Dens Deck Prime Minimum ¼-inch thick	**9, 10, 11 with 13**	**1:2.7 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –45 psf (See General Limitation #9)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(42): All layers of insulation simultaneously attached; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1-inch thick	**N/A**	**N/A**

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Dens Deck Prime Minimum ½-inch thick	**1, 2, with 5**	**1:1.66 ft²**
Plywood Minimum 19/32-inch thick	**1, 2, with 5**	**1:1.66 ft²**

Note: All layers of insulation fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Membrane: Sure-Weld FleeceBACK membrane adhered to the insulation with HydroBond Water-Based Bonding Adhesive. The adhesive shall be applied to the substrate only at a rate of 120 ft²/gallon. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –52.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced, FleeceBACK

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(43): Base layer(s) of insulation fastened, top layer adhered; membrane adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Insulfoam EPS Minimum 3-inch thick	**1, 2, 4 with 5**	**1:2 ft²**

Note: The base layer of insulation is fastened into the steel deck. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment.

Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Insulfoam Insul-Lam Minimum 2-inch thick	**N/A**	**N/A**

Note: Top insulation layer is adhered with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank Adhesives applied to the base layer in approximately ¾ to 1-inch wide beads spaced 6-inches on center.

Membrane: Sure-Weld, Sure-Weld EXTRA or Sure-Weld HS membrane adhered to the insulation using Sure-Weld Bonding Adhesive or Low VOC Bonding Adhesive. The adhesive is applied in a contact application and is applied to both the underside of the roofing membrane and the top side of the approved substrate at a rate of 60 ft²/gallon, finished surface area. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.
OR
Sure-Weld FleeceBACK membrane adhered to the insulation with Flexible FAST, Flexible FAST Dual Cartridge or Flexible FAST 5-Gallon Jug Adhesive applied to the substrate in approximately ¾ to 1-inch wide beads spaced 6-inches on center. Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld.

Maximum Design Pressure: –75 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW #12 HWH Teks/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW #10 HWH Teks/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(1): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Note: Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or FAST Dual Tank, Carlisle Olybond 500BA, Carlisle One-Step Adhesive Millennium One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening Option #1: Mechanically fastened 12-inches on center through the field seams rows 72-inches on center using HPX Fasteners and Piranha Plates.
Maximum Design Pressure: -45psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW #12 HWH Teks/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW #10 HWH Teks/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(2): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Note: Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or FAST Dual Tank, Carlisle Olybond 500BA, Carlisle One-Step Adhesive Millennium One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(4): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 12-inches on center through the field seams using HPX Fasteners and Piranha Plates rows spaced 10-feet on center.

Maximum Design Pressure: -45 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 66-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(5): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 12-inches on center through the field seams using HPX Fasteners and Piranha Plates rows spaced 10-feet on center.

Maximum Design Pressure: -45 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 20 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(6): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 60-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(7): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 12-inches on center through the field seams using HPX Fasteners and Piranha Plates rows spaced 10-feet on center.

Maximum Design Pressure: -45 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 60-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(8): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG , H-Shield HD Composite Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(9): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 6-inches on center through the field seams using HPX Fasteners and Piranha Plates rows spaced 8-feet on center.

Maximum Design Pressure: -60 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 66-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(10): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 54-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(13): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 6-inches on center through the field seams using HPX Fasteners and Piranha Plates rows spaced 12-feet on center

Maximum Design Pressure: -60 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 60-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 30 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(15): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite Minimum 1.2-inch thick		
SecurShield HD or H-Shield HD Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime Minimum ¼-inch thick	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga. to 20 ga, Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 60-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(16): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Base and/or Top Insulation Layer **InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite** **Minimum 1.2-inch thick**		
SecurShield HD or H-Shield HD **Minimum ½-inch thick**	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board **Minimum ¼-inch thick**	**N/A**	**N/A**
Dens Deck, Dens Deck Prime **Minimum ¼-inch thick**	**N/A**	**N/A**

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(17): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 12-inches on center through the field seams using HPX Fasteners and Piranha Plates or Trufast #15 EHD Fastener and Trufast 2.4-inch Barbed Metal Seam Plate rows spaced 8-feet on center

Maximum Design Pressure: -45 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 16 -18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 60-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 20 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 18-20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 66-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(18): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 60-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(19): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18-22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(20): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank, Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 12-inches on center through the field seams using HPX Fasteners and Piranha Plates or Trufast #15 EHD Fastener and Trufast 2.4-inch Barbed Metal Seam Plate rows spaced 4-feet on center

Maximum Design Pressure: -52.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 20 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(21): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 66-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 48-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(22): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank, Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 20 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 54-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(23): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank, Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(24): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank, Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 6-inches on center through the field seams using HPX Fasteners and Piranha Plates or Trufast #15 EHD Fastener and Trufast 2.4-inch Barbed Metal Seam Plate rows spaced 8-feet on center.

Maximum Design Pressure: -60.0 psf. (See General Limitation #7*)*

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 to 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 60-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(25): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(26): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive Flexible FAST Dual Tank Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 6-inches on center through the field seams using HPX Fasteners and Piranha Plates or Trufast #15 EHD Fastener and Trufast 2.4-inch Barbed Metal Seam Plate rows spaced 4-feet on center

Maximum Design Pressure: -75 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18-22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.
Or
Minimum 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 60-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(27): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive Flexible FAST Dual Tank Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(28): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.2-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation and base sheet shall be simultaneously attached. See base sheet below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with Carlisle Flexible FAST, Flexible FAST Dual Cartridge, Flexible FAST 5-Gallon Jug Adhesive or Flexible FAST Dual Tank, Carlisle Olybond 500BA, Millennium One Step Foamable Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Membrane: Sure-Weld, Sure-Weld HS or Sure-Weld EXTRA, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: Fastened 6-inches on center through the field seams using HPX Fasteners and Piranha Plates or Trufast #15 EHD Fastener and Trufast 2.4-inch Barbed Metal Seam Plate rows spaced 4-feet on center

Maximum Design Pressure: -112.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(29): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft.

Thermal Barrier: (Optional) Thermal barrier of any approved type or thickness, loose-laid, adhered or mechanically attached.

Membrane: Minimum 60-mil Sure-Weld or Sure-Weld EXTRA membrane secured through the preliminarily attached insulation as specified below.

Fastening: 10-foot wide Sure-Weld, or Sure-Weld EXTRA membrane mechanically fastened 6-inches on center through the field seams using HPX Fasteners and Piranha Plates

Membrane Splices: Overlap membrane splices a minimum of 5-1/2 inches to provide for a minimum 1-1/2 inch heat weld along the outside edge.

Maximum Design Pressure: -52.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 16 - 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 24 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(30): Membrane induction welded over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements.

Membrane: Sure-Weld or Sure-Weld EXTRA membrane induction welded over mechanically fastened insulation.

Fastening: Attached with HPX Fasteners and RhinoBond Plates-TPO at the minimum rate of 1:5.33 ft² (6 per 4 ft by 8 ft board).

Maximum Design Pressure: -45 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 - 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(31): Membrane induction welded over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements.

Membrane: Sure-Weld or Sure-Weld EXTRA membrane induction welded over mechanically fastened insulation.

Fastening: Attached with HPX Fasteners and RhinoBond Plates-TPO at the minimum rate of 1:4 ft² (8 per 4 ft by 8 ft board).

Maximum Design Pressure: -52.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 - 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(32): Membrane induction welded over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements.

Membrane: Sure-Weld or Sure-Weld EXTRA membrane induction welded over mechanically fastened insulation.

Fastening: Attached with HPX Fasteners and RhinoBond Plates-TPO at the minimum rate of 1:4 ft² (8 per 4 ft by 8 ft board).

Maximum Design Pressure: -52.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 - 22 ga., Type B, Grade 33 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(33): Membrane induction welded over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements.

Membrane: Sure-Weld or Sure-Weld EXTRA membrane induction welded over mechanically fastened insulation.

Fastening: HPX Fasteners and RhinoBond Plates-TPO at the minimum rate of 1:2 ft² (16 per 4 ft by 8 ft board).

Maximum Design Pressure: -82.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 - 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(34): Membrane induction welded over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements.

Membrane: Sure-Weld or Sure-Weld EXTRA membrane induction welded over mechanically fastened insulation.

Fastening: HPX Fasteners and RhinoBond Plates-TPO at the minimum rate of 1:2 ft² (16 per 4 ft by 8 ft board).

Maximum Design Pressure: -97.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 16 - 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex #10 HWH Teks 1 fasteners at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(35): Membrane induction welded over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements.

Membrane: Sure-Weld or Sure-Weld EXTRA membrane induction welded over mechanically fastened insulation.

Thermal Barrier: (Optional) Thermal barrier of any approved type or thickness, loose-laid, adhered or mechanically attached.

Membrane Assembly: RhinoBond Plate and HPX Fasteners are installed 12-inches on center, in rows spaced maximum 5-feet on center. Minimum 60-mil Sure-Weld or Sure-Weld EXTRA membrane induction welded to RhinoBond Plates-TPO using RhinoBond induction tool.

Membrane Splices: Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld along the outside edge.

Maximum Design Pressure: -52.5 psf. (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.
Or
Minimum 18 - 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.
Or
Minimum 18 - 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 54-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(36): Membrane induction welded over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements.

Thermal Barrier: (Optional) Thermal barrier of any approved type or thickness, loose-laid, adhered or mechanically attached.

Membrane Assembly: RhinoBond Plate and HPX Fasteners are installed 6-inches on center, in rows spaced maximum 10-feet on center. Minimum 60-mil Sure-Weld or Sure-Weld EXTRA membrane induction welded to RhinoBond Plates-TPO using RhinoBond induction tool.

Membrane Type: Single Ply, Thermoplastic, TPO, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 18 - 22 ga., Type B, Grade 80 steel deck secured to structural steel supports spaced a maximum 72-inches on center with ITW Buildex Traxx/5 fasteners at a maximum of 6 inches on center. Deck side laps are secured with ITW Buildex Traxx/1 at a maximum spacing of 12 inches on center.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(37): Membrane induction welded over mechanically fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base and/or Top Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
InsulBase, SecurShield, SecurShield HD Composite, H-Shield, H-Shield CG, H-Shield HD Composite		
Minimum 1.5-inch thick	**N/A**	**N/A**
SecurShield HD or H-Shield HD		
Minimum ½-inch thick	**N/A**	**N/A**
SECUROCK Gypsum Fiber or Glass Mat Roof Board		
Minimum ¼-inch thick	**N/A**	**N/A**
Dens Deck, Dens Deck Prime		
Minimum ¼-inch thick	**N/A**	**N/A**

Note: All layers of insulation shall be simultaneously attached. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements.

Thermal Barrier: (Optional) Thermal barrier of any approved type or thickness, loose-laid, adhered or mechanically attached.

Membrane Assembly: RhinoBond Plate and HPX Fasteners are installed 6-inches on center, in rows spaced maximum 5-feet on center. Minimum 60-mil Sure-Weld or Sure-Weld EXTRA membrane induction welded to RhinoBond Plates-TPO using RhinoBond induction tool.

Membrane Splices: Overlap membrane splices a minimum of 2 inches to provide for a minimum 1-1/2 inch heat weld along the outside edge.

Maximum Design Pressure: -67.5 psf. (See General Limitation #7)